CO_2-EOR DIGITAL TWIN

Delaware Basin CO_2 Enhanced Oil Recovery Command Center

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

NEAR REAL-TIME DIGITAL TWIN & MULTI-AGENT AI FOR CO_2-EOR FIELD OPERATIONS

CO_2-EOR Digital Twin delivers a unified command center fusing geospatial field monitoring, subsurface fiber optic sensing (DAS/DTS), multi-agent AI advisory, injection pattern optimization, and carbon accounting on the Databricks Lakehouse Platform.

◆ BUSINESS VALUE

- **Maximize oil recovery** with AI-optimized CO_2 injection patterns and WAG ratio tuning
- **Track carbon storage compliance** with real-time CO_2 balance and accounting dashboards
- **Reduce subsurface uncertainty** with integrated DAS/DTS fiber optic and microseismic monitoring
- **Improve field economics** with per-well revenue, OpEx, and netback analysis
- **Accelerate decision-making** through 5 specialized AI agents providing contextual recommendations

◆ CHALLENGES ADDRESSED

- **Fragmented field data** — SCADA, reservoir models, and economic tools in disconnected systems
- **Limited subsurface visibility** — no integrated view of DAS, DTS, and microseismic data
- **Manual carbon accounting** — spreadsheet-based CO_2 tracking prone to errors
- **Reactive pattern management** — injection adjustments based on lagging monthly reports
- **Siloed expertise** — reservoir, operations, and commercial teams working independently

◆ WHO IT'S FOR

Reservoir Engineer
Subsurface monitoring, saturation maps, pressure analysis, and pattern optimization

Operations Manager
Real-time field overview, well status, and operational KPI tracking

Carbon/ESG Analyst
CO_2 balance tracking, storage compliance, and emissions reporting

Commercial Analyst
Per-well economics, netback analysis, and revenue forecasting

24	5	8	4
Wells Monitored	AI Agents	Dashboard Tabs	Flood Patterns

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the CO_2-EOR Digital Twin repo to your local machine
2. **Build Frontend**
   Run npm install & npm run build for the React UI
3. **Deploy the App**
   Run databricks apps deploy to launch the command center
4. **Monitor & Explore**
   Open the app URL and explore all 8 tabs

◆ GETTING THE MOST OUT OF CO2-EOR DIGITAL TWIN

Explore the Field Map
Use the MapLibre GL geospatial overview to monitor wells, pipelines, facilities, and fleet vehicles

Consult AI Agents
Chat with 5 specialized agents for monitoring, optimization, maintenance, and commercial insights

Dive into Subsurface
Analyze DAS waterfall plots, DTS profiles, reservoir saturation maps, and microseismic events

Track Carbon Balance
Monitor injected, recycled, purchased, and stored CO_2 volumes for compliance

◆ TARGET INDUSTRY

Upstream Oil & Gas — CO_2 Enhanced Oil Recovery operations, mature field rejuvenation, carbon capture utilization & storage (CCUS), and tertiary recovery programs in the Permian and Delaware Basins.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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